

Namal Blood Bank Management System (NBBMS)

Database Design Document

V 3.0

By

Group Members

Roll Numbers

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30/6/24	V.3.0	Improved ERD (self explanatory)	Mam Asiya Batool
10/6/24	V.2.0	Removed extra attributes like Group Blood Type, Status, expiry date derived attribute.	Mam Asiya Batool
22/04/24	V 1.0	No changes	Mam Asiya Batool

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CHAPTER 1: PROJECT OVERVIEW

1.1. INTRODUCTION:

Our project aims to develop a comprehensive Namal Blood Bank Management System (NBBMS) utilizing MySQL for database management and Python/JavaScript for backend logic along with HTML & CSS for frontend development. The system addresses the critical need for efficient management of blood donation, storage, and distribution processes within blood banks.

In a world where efficient blood management is paramount, the NBBMS addresses the critical need for accurate record-keeping of donors, blood units, and recipients. By streamlining processes and minimizing manual data entry errors, the system aims to overcome challenges such as inefficient blood tracking and delays in matching donor blood types with recipient needs.

The ultimate goal of the NBBMS is to provide blood bank staff with a powerful toolset for managing donor registration, donation recording, blood unit storage, recipient registration, and transfusion management. By enhancing data accuracy, improving tracking capabilities, and enabling timely matching of donor blood types with recipient needs, the NBBMS ensures that blood banks can operate with maximum efficiency and effectiveness.

1.2. PROBLEM STATEMENT:

The NBBMS addresses the challenges faced by Blood Wing of NSSI Society of Namal University in maintaining accurate records of donors, blood units, networks (blood Banks) and recipients. These challenges include manual data entry errors, inefficient blood tracking, and difficulties in matching donor blood types with recipient needs in a timely manner. By implementing the NBBMS, we aim to mitigate these issues and improve the overall management of blood donation, storage, and distribution processes.

1.3. PROJECT OBJECTIVES:

- Design and implement an efficient database schema for storing donor, donation, blood bag, recipient, and related data.
- Develop user-friendly interfaces for blood bank staff to input and access relevant information.
- Implement functionalities for managing blood units from donation to transfusion, including expiry management.
- Enable efficient search and matching of donor blood types with recipient needs.
- Ensure data security and integrity through appropriate access controls and validation mechanisms.

1.4. DOCUMENT OBJECTIVES:

The objectives of this documentation are to provide a detailed overview of the database design and implementation progress achieved in the fourth milestone of the NBBMS project. It aims to define and describe entities, establish relationships, create an Entity Relationship Diagram (ERD), design a relational schema, and demonstrate the practical use of the database through real time data and SQL queries. The documentation also ensures proper citation of all sources consulted.

CHAPTER 2: DETAILED DATABASE DESIGN

2.1. ENTITY:

The entities are given below:

Sr. No	Entity Name	Description
01	STAFF	The STAFF entity represents the individuals working within the blood bank facility who are responsible for managing various aspects of blood donation, storage, and distribution processes.
02	DONOR	The DONOR entity represents individuals who voluntarily want to donate blood to the blood bank for use in transfusions.
03	RECIPIENT	The RECIPIENT entity represents individuals who require blood transfusions due to medical reasons.
04	BLOOD_BAGS	The BLOOD-BAGS entity is an “ associative entity ” which is intended to manage information related to blood bags used for storing donated blood.
05	NETWORKS	The NETWORKS entity represents connections or interactions between the blood bank and other healthcare facilities or organizations.

2.2. DATA DICTIONARY:

The entities and their attributes are given below:

2.2.1. STAFF:

Sr. No	Name	Data Type	Constraint	Description
01	<u>S_ID</u>	INT	NOT NULL, AUTO_INCREMENT, PRIMARY KEY	It uniquely identifies each staff member within the blood bank system.
02	Name	CHAR(50)	NOT NULL	It stores the name of the staff member.
03	Contact	CHAR(12)	NOT NULL	It holds the contact number of the staff member.
04	CNIC	CHAR(15)	NOT NULL	It stores the National Identity Card (CNIC) number of the staff member.
05	Address	CHAR(50)	No constraint	It represents the address of the staff member.

2.2.2. DONOR:

Sr. No	Name	Data Type	Constraint	Description
01	<u>D_CNIC</u>	CHAR(15)	NOT NULL, PRIMARY KEY	It uniquely identifies each donor within the blood bank system using their CNIC.
02	Name	CHAR(50)	NOT NULL	It stores the name of the donor.
03	Contact	CHAR(12)	NOT NULL	It holds the contact number of Donor.
04	Department	CHAR(15)	No constraint	It represents the department or field of work of the donor.
05	Category	CHAR(15)	No constraint	It categorizes the donor based on position (Student, Staff, or Faculty of Namal)
06	Blood_Group	CHAR(3)	NOT NULL	It stores the blood group of the donor.

2.2.3. RECIPIENT:

Sr. No	Name	Data Type	Constraint	Description
01	<u>R_CNIC</u>	CHAR (15)	NOT NULL, PRIMARY KEY	It uniquely identifies each recipient within the blood bank system using their CNIC.
02	Name	CHAR(50)	NOT NULL	It stores the name of the recipient.
03	Contact	CHAR(12)	NOT NULL	It holds the contact number of the recipient.
04	Address	VARCHAR(100)	No constraint	It represents the address of the recipient.
05	Institute	VARCHAR(100)	No constraint	It stores the institution or medical facility associated with the recipient.

2.2.4. BLOOD-BAGS:

Sr. No	Name	Data Type	Constraint	Description
01	<u>Bag_ID</u>	INT	NOT NULL, AUTO_INCREMENT, PRIMARY KEY	It uniquely identifies each staff member within the blood bank system. It uniquely identifies each blood bag within the blood bank system.
03	Donation_Date	Date	NOT NULL	It records the date when the blood was donated and the bag was created.

05	Delivery_Date	Date	No constraint	It indicates the date when the blood bag was delivered for transfusion or other purposes.
----	---------------	------	---------------	---

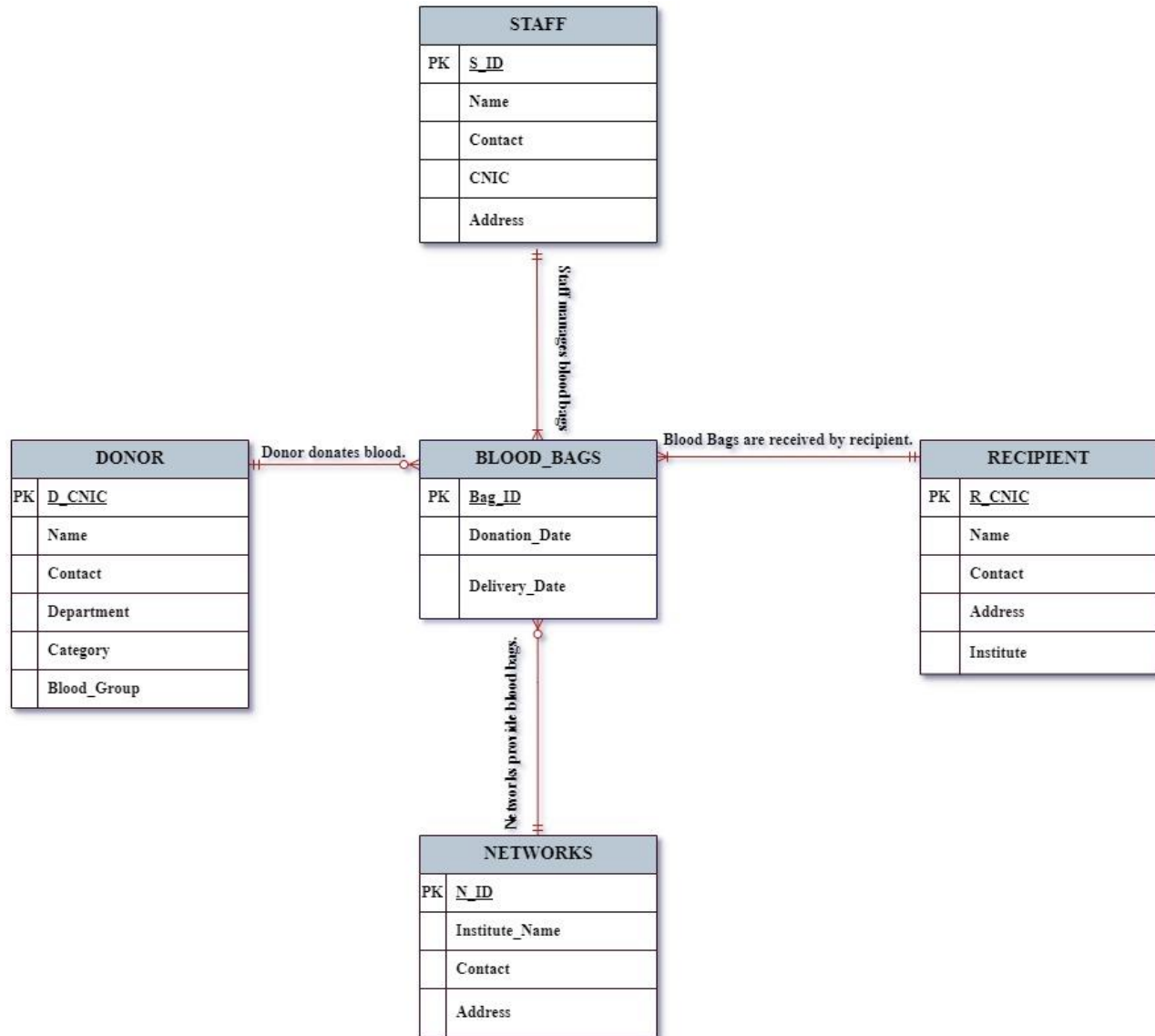
2.2.5. NETWORKS:

Sr. No	Name	Data Type	Constraint	Description
01	<u>N_ID</u>	INT	NOT NULL, AUTO_INCREMENT, PRIMARY KEY	It uniquely identifies each networking connection within the blood bank system.
02	Institute_Name	VARCHAR(100)	NOT NULL	It stores the name of the institution or organization for networking purposes.
03	Contact	CHAR(13)	NOT NULL	It holds the contact number of the networking connection.
04	Address	VARCHAR(100)	No constraint	It represents the address of the institution or organization for networking.

2.3. RELATIONSHIPS:

Sr. No	Participating Entities	Relation	Business Rule
01	DONOR, BLOOD_BAGS	Donor donates blood. One Mandatory to Many Optional.	One donor may donate many blood bag. One blood bag may donated by one donor.
02	RECIPIENT, BLOOD_BAGS	Recipient receives blood bags. Many Mandatory to One Mandatory.	One blood bag can be donated to one recipient. One recipient can receive many blood bags.
03	STAFF, BLOOD_BAGS	Staff manages blood bags and its Information. One Mandatory to Many One.	One staff may manage many blood bags. One blood bag is managed by one staff.
04	NETWORKS, BLOOD_BAGS	Networks provide. blood bags. One Mandatory to Many Optional.	One network may donate many blood bag. One blood bag may donated by one network.

2.4. ENTITY RELATIONSHIP DIAGRAM:

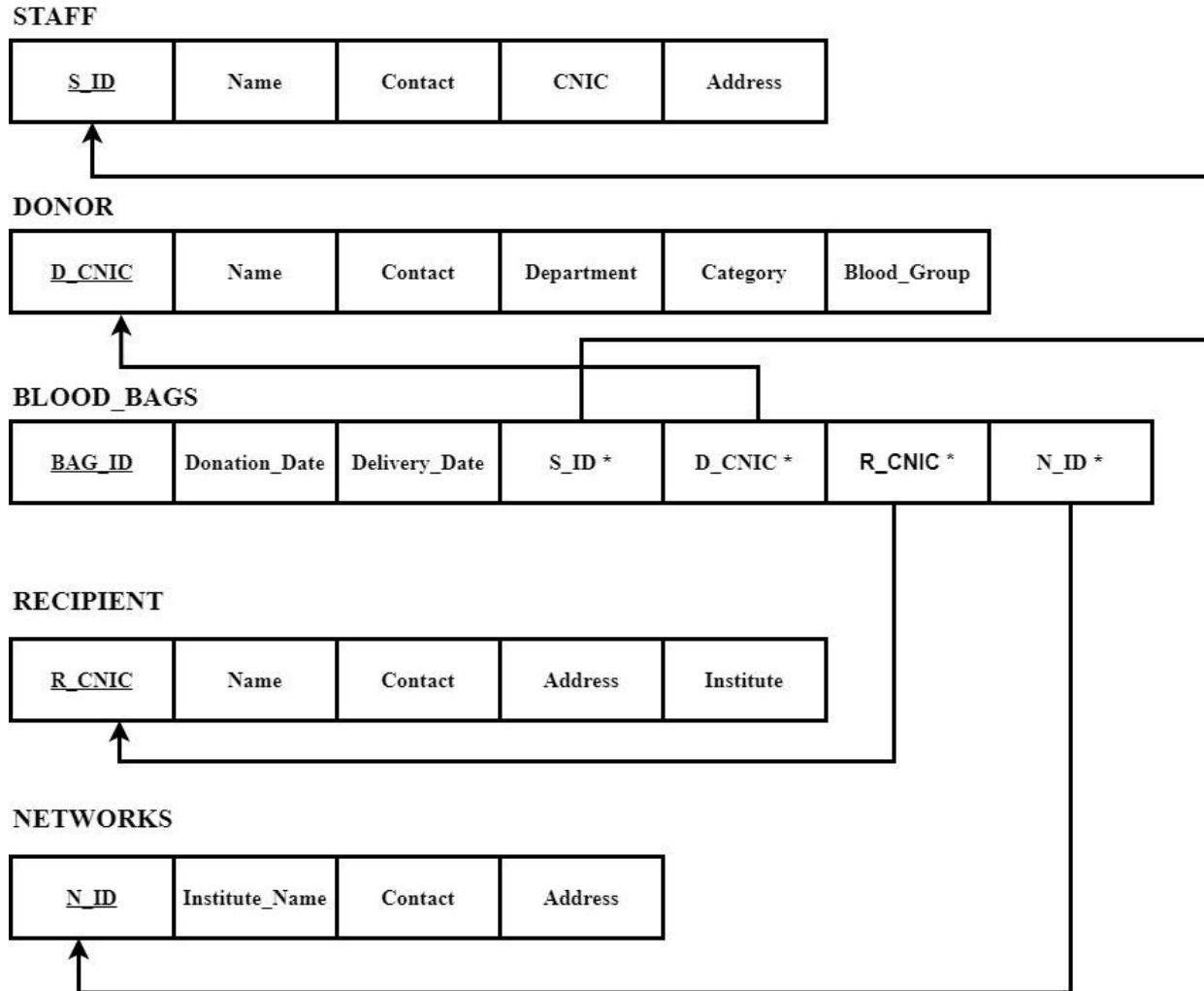


ERD of Namal Blood Bank Management System (NBBMS) [1] [2]

CHAPTER 3 : LOGICAL DATABASE DESIGN

3.1. RELATIONAL SCHEMA:

Relational Schema of “*Namal Blood Bank Management System (NBBMS)*” is given below: [1]



3.2. FUNCTIONAL DEPENDENCIES:

STAFF

- S_ID → Name, Contact, CNIC, Address

Example:

If S_ID = 13, then Name = Umair Ahmad, Contact = 0326-1218644

, CNIC = 32101-7676321-4, Address = Namal University Mianwali

DONOR

- **D_CNIC → Name, Contact, Department, Category, Blood_Group**

Example:

- If D_CNIC = 30007-0000007-7, then Name = said Anwar, Contact = 0302-6065103, Department = CS, Category = 4th Year , Blood_Group = A+

RECIPIENT

- **R_CNIC → Name, Contact, Address, Institute**

Example:

If R_CNIC = 32919-3698521-1, then Name = Sadia, Contact = 0339-0123456, Address = Ban Hafiz Jee ,Mianwali, Institute = Tasneem hospital, bun

BLOOD-BAGS

- **Bag_ID → Donation_Date, Delivery_Date, D_CNIC (Donor), R_CNIC (Recipient), S_ID (Staff)**
- **Bag_ID → Donation_Date, Delivery_Date**

Example:

- If Bag_ID = 7, then Donation_Date = 2024-05-29, Delivery_Date = 2024-06-03, D_CNIC = 30336-0000336-3, R_CNIC = 32737-7654321-9, S_ID = 5

NETWORKS

- **N_ID → Institute_Name, Contact, Address**

Example:

- If N_ID = 5, then Institute_Name = Agha Khan Blood Bank Mianwali, Contact = (0459) 227589, Address = Near Agha Khan Diagnostic Center, Mianwali

3.3. NORMALIZATION:

3.3.1. First Normal Form (1NF)

- Ensure each table has a primary key.
- Ensure no repeating groups or arrays.

All provided tables are in 1NF as each table has a primary key and contains atomic values.

3.3.2. Second Normal Form (2NF)

- Achieve 1NF.
- Remove partial dependencies (i.e., no non-prime attribute should depend on a part of a candidate key).

Relations:

STAFF

- Already in 2NF because S_ID is the primary key and all attributes depend on it fully.

DONOR

- Already in 2NF because D_CNIC is the primary key and all attributes depend on it fully.

RECIPIENT

- Already in 2NF because R_CNIC is the primary key and all attributes depend on it fully.

BLOOD-BAGS

- Already in 2NF because Bag_ID is the primary key and all attributes depend on it fully.

NETWORKS

- Already in 2NF because N_ID is the primary key and all attributes depend on it fully.

3.3.3. Third Normal Form (3NF)

- Achieve 2NF.
- Remove transitive dependencies.

Relations:

STAFF

- Already in 3NF because there are no transitive dependencies.
- **S_ID → Name, Contact, CNIC, Address**

DONOR

- Already in 3NF because there are no transitive dependencies.
- **D_CNIC → Name, Contact, Department, Category, Blood_Group**

RECIPIENT

- Already in 3NF because there are no transitive dependencies.
- **R_CNIC → Name, Contact, Address, Institute**

BLOOD-BAGS

- Already in 3NF because there are no transitive dependencies.
- **Bag_ID → Donation_Date, Delivery_Date**

NETWORKS

- Already in 3NF because there are no transitive dependencies.

N_ID → Institute_Name, Contact, Address

CHAPTER 4 : PHYSICAL DATABASE DESIGN

4.1. STRUCTURE OF THE TABLES: [2]

STAFF:

```
mysql> DESCRIBE STAFF;
```

Field	Type	Null	Key	Default	Extra
S_ID	int	NO	PRI	NULL	auto_increment
NAME	char(50)	NO		NULL	
CONTACT	char(12)	NO		NULL	
CNIC	char(15)	NO		NULL	
ADDRESS	varchar(50)	YES		NULL	

5 rows in set (0.00 sec)

DONOR:

```
mysql> DESCRIBE donor;
```

Field	Type	Null	Key	Default	Extra
D_CNIC	char(15)	NO	PRI	NULL	
Name	char(50)	NO		NULL	
Contact	char(12)	NO		NULL	
Department	char(15)	YES		NULL	
Category	char(15)	YES		NULL	
Blood_Group	char(3)	NO		NULL	

6 rows in set (0.00 sec)

RECIPIENT:

```
mysql> DESCRIBE recipient;
```

Field	Type	Null	Key	Default	Extra
R_CNIC	char(15)	NO	PRI	NULL	
NAME	char(50)	NO		NULL	
CONTACT	char(12)	NO		NULL	
ADDRESS	varchar(100)	YES		NULL	
INSTITUTE	varchar(100)	YES		NULL	

5 rows in set (0.00 sec)

NETWORKS:

```
mysql> DESCRIBE networks;
```

Field	Type	Null	Key	Default	Extra
N_ID	int	NO	PRI	NULL	auto_increment
Institute_Name	varchar(100)	NO		NULL	
Contact	char(13)	NO		NULL	
Address	varchar(100)	YES		NULL	

```
4 rows in set (0.00 sec)
```

BLOOD_BAGS:

```
mysql> DESCRIBE blood_bags;
```

Field	Type	Null	Key	Default	Extra
Bag_ID	int	NO	PRI	NULL	auto_increment
Donation_Date	date	YES		NULL	
Delivery_Date	date	YES		NULL	
S_ID	int	YES	MUL	NULL	
D_CNIC	char(15)	YES	MUL	NULL	
R_CNIC	char(15)	YES	MUL	NULL	
N_ID	int	YES	MUL	NULL	

```
7 rows in set (0.00 sec)
```

4.2. DATA SAMPLES INSIDE TABLES:

SELECT * FROM Staff;

S_ID	NAME	CONTACT	CNIC	ADDRESS
1	Shazam	0321-1234567	32101-7654321-2	Namal University Mianwali
2	Abdul Raheem	0321-1234568	32101-7654321-3	Namal University Mianwali
3	Rehan	0321-1234569	32101-7654321-4	Namal University Mianwali
4	Awais	0321-1234570	32101-7654321-5	Namal University Mianwali
5	Ghazia	0321-1234571	32101-7654321-6	Namal University Mianwali
6	Ali Taimoor	0321-1234572	32101-7654321-7	Namal University Mianwali
7	Anila	0321-1234573	32101-7654321-8	Namal University Mianwali
8	Nouman	0321-1234574	32101-7654321-9	Namal University Mianwali
9	Shariq Khan	0321-1234575	32101-7654321-1	Namal University Mianwali
10	Umar Shehzad	0321-1234576	32101-7654321-1	Namal University Mianwali
11	Zoya	0321-1234577	32101-7654321-6	Namal University Mianwali
12	Shahid	0321-1234578	32101-7654321-1	Namal University Mianwali
13	Umair Ahmed	0321-1234579	32101-7676321-4	Namal University Mianwali
14	Kamran Malik	0321-1234580	32101-7657321-5	Namal University Mianwali
15	Asim Khan	0321-1234581	32101-7666321-8	Namal University Mianwali
16	Noor Fatima	0321-1234582	32101-7654351-6	Namal University Mianwali

```
16 rows in set (0.00 sec)
```

SELECT * FROM Donor;

Namal Blood Bank Management System (NBBMS)

D_CNIC	Name	Contact	Department	Category	Blood_Group
30005-0000005-5	M Saleem	0301-2685998	NULL	Faculty	A+
30014-0000014-1	M Irfan	0335-6302675	CS	3rd Year	A+
30018-0000018-5	Muhammad Usman	0345-6443752	EE	2nd Year	O+
30033-0000033-6	muhammad Usman	0301-7720324	EE	3rd Year	A+
30047-0000047-2	Aqsa	0311-7121603	EE	1st Year	A+
30049-0000049-4	sana Malik	0324-5972465	BBA	3rd Year	A+
30069-0000069-6	Ahad Khan	0309-6556446	NULL	2nd Year	A+
30076-0000076-2	Khan Mehmood	0307-6566654	NULL	Faculty	A+
30092-0000092-2	Sana	0349-6084322	EE	1st Year	A+
30104-0000104-5	Siuuira	0301-5840522	CS	1st Year	A+
30134-0000134-8	riaz u din	0346-0520951	EE	3rd Year	AB+
30164-0000164-2	Quratulain	0306-8454860	MATHS	1st Year	AB-
30167-0000167-5	Nisar Ali	0345-4779109	NULL	Faculty	B+
30214-0000214-6	Naseem	0347-7611603	Math	2nd Year	O+
30220-0000220-4	Bisma amir	0323-1897256	Math	3rd Year	B+
30524-0000524-2	Sadia	0302-7178430	BBA	2nd Year	B+
30528-0000528-6	Ibrahim	0347-7373032	EE	1st Year	B+
30529-0000529-7	Shazia Nazir	0342-6871795	CS	2nd Year	O+
30534-0000534-0	Nida Anwar	0308-7981364	Math	2nd Year	A+
30535-0000535-1	Shahbaz	0343-7743305	CS	3rd Year	B+
30543-0000543-4	Salma	0305-6291897	CS	1st Year	O+
30547-0000547-3	Naeem	0342-9574186	CS	3rd Year	A+
30550-0000550-5	Aziz Ul Haq	0321-9451601	CS	1st Year	O+
30553-0000553-9	Bushra	0311-9982304	CS	1st Year	B+
30562-0000562-4	Wazir M	0333-7363017	EE	1st Year	O+
30568-0000568-1	basit	0327-0367753	EE	1st Year	O+
30569-0000569-2	Aqsa	0327-3299072	MATHS	1st Year	O+
30573-0000573-6	Rehana	0325-8174066	MATHS	1st Year	O+

52 rows in set (0.00 sec)

SELECT * FROM Recipient;

R_CNIC	NAME	CONTACT	ADDRESS	INSTITUTE
30535-3698741-7	Aiza	0355-4321098	Agha khan center, Minawali	Rehman Hospital, Mianwali
30545-7654321-7	Zoya	0365-8901234	HGJV+639, Mianwali	Obaid Noor Hospital, Mianwali
31626-2468013-8	Javed	0346-7890123	Civil Hospital, Mianwali	Blood Bank DHQ Hospital Mianwali
31636-1234567-8	Saad	0356-9012345	Gulberg Chock, Minawali	Namal hospital
31646-2468013-8	Raza	0366-7890123	Civil Hospital, Mianwali	Rehman Hospital, Mianwali
31818-2468013-0	Umar	0338-5432109	Dhurnaka, Minawali	Rehman Hospital, Mianwali
31909-7654321-1	Saira	0329-5678901	Gulberg Chock, Minawali	Tabish Children Hospital, Mianwali
32010-2468013-2	Arman	0330-1098765	Gulberg Chock, Minawali	Rehman Hospital, Mianwali
32101-7654321-2	Adnan	0321-1234567	Ballo khail RD, Minawali	Rehman Hospital, Mianwali
37323-3698741-5	Khadija	0343-2345678	Gulberg Chock, Minawali	Agha Khan Blozod Bank Mianwali
37515-3698741-7	Maryam	0335-8901234	Agha khan center, Minawali	Sarmad Trust Blood Bank
37606-2468135-8	Asad	0326-9876543	Gulberg Chock, Minawali	Kashaf Homoeopathic Clinic, Mianwali
38333-7890123-5	Daniyal	0353-3210987	Main Street, Minawali	Tasneem hospital, bun
38343-3698741-5	Humna	0363-8765432	Gulberg Chock, Minawali	Sarmad Trust Blood Bank
38424-1234567-6	Adil	0344-8765432	Gulberg Chock, Minawali	Obaid Noor Hospital, Mianwali
38616-1234567-8	Waseem	0336-4321098	Gulberg Chock, Minawali	Agha Khan Blozod Bank Mianwali
38707-3698741-9	Amina	0327-4567890	Namal, Minawali	Rehman Hospital, Mianwali
39434-2468135-6	Aiman	0354-8901234	Near Civil Hospital, Minawali	Kashaf Homoeopathic Clinic, Mianwali
39444-1234567-6	Aqsa	0364-3456789	Gulberg Chock, Minawali	Agha Khan Blozod Bank Mianwali
39525-7654321-7	Rukhsana	0345-3456789	HGJV+639, Mianwali	Obaid Noor Hospital, Mianwali
39717-7654321-9	Ayesha	0337-9012345	Gulberg Chowk, Minawali	Obaid Noor Hospital, Mianwali
39808-1234567-0	Faisal	0328-0987654	Ballo khail RD, Minawali	Namal hospital

49 rows in set (0.00 sec)

SELECT * FROM Networks;

N_ID	Institute_Name	Contact	Address
1	Blood Bank DHQ Hospital Mianwali	(0459) 235136	HGJV+639, Mianwali, Punjab
2	Naml Wellness Blood Bank	(0459) 229847	Naml College Road, Mianwali
3	Red Crescent Blood Bank Mianwali	(0459) 228035	Main Street, Mianwali
4	Sarmad Trust Blood Bank	(0459) 226043	Near Civil Hospital, Mianwali
5	Agha Khan Blood Bank Mianwali	(0459) 227589	Near Agha Khan Diagnostic Center, Mianwali

5 rows in set (0.00 sec)

SELECT * FROM Blood_Bags;

Bag_ID	Donation_Date	Delivery_Date	S_ID	D_CNIC	R_CNIC	N_ID
1	2024-05-20	2024-05-30	1	30005-0000005-5	33848-1234567-0	NULL
2	2024-05-21	2024-05-22	2	30033-0000033-6	34929-7654321-1	NULL
3	2024-05-23	NULL	3	30134-0000134-8	NULL	NULL
4	2024-05-24	2024-06-03	4	NULL	36222-2468135-4	3
5	2024-05-24	2024-06-01	5	30316-0000316-1	37515-3698741-7	NULL
6	2024-05-27	NULL	6	NULL	NULL	5
7	2024-05-29	2024-06-03	5	30336-0000336-3	32737-7654321-9	NULL
8	2024-05-31	NULL	3	30347-0000347-5	NULL	NULL
9	2024-06-01	NULL	2	30255-0000255-3	NULL	NULL
10	2024-06-01	2024-06-22	9	NULL	31818-2468013-0	1
11	2024-06-01	2024-06-20	7	NULL	39808-1234567-0	1
12	2024-06-02	2024-06-05	2	30507-0000507-2	33838-2468013-0	NULL
13	2024-06-03	2024-06-08	8	NULL	32101-7654321-2	4
14	2024-06-03	2024-06-12	6	30553-0000553-9	36505-7890123-7	NULL
15	2024-06-04	2024-06-10	4	NULL	34212-1357902-4	2

15 rows in set (0.00 sec)

4.3. QUERIES RESULTS:

Queries for Namal Blood Bank Database											
1	Query for counting total no of unique blood groups.										
Query	SELECT Blood_Group, COUNT(*) AS NumberOfDonors FROM DONOR GROUP BY Blood_Group HAVING COUNT(*) > 1 ORDER BY NumberOfDonors DESC;										
Output	<table> <tr> <th>Blood_Group</th><th>NumberOfDonors</th></tr> <tr> <td>B+</td><td>20</td></tr> <tr> <td>A+</td><td>16</td></tr> <tr> <td>O+</td><td>12</td></tr> <tr> <td>B-</td><td>2</td></tr> </table> 4 rows in set (0.01 sec)	Blood_Group	NumberOfDonors	B+	20	A+	16	O+	12	B-	2
Blood_Group	NumberOfDonors										
B+	20										
A+	16										
O+	12										
B-	2										
2	This query retrieves the names, contact information, blood groups, and donation counts of										

	donors with the blood group 'A+' .In a blood_bags tables and shows the results by the donor's details, and sorts them by the number of donations in descending order.
Query	<pre>SELECT D.Name, D.Contact, D.Blood_Group, COUNT(BB.Bag_ID) AS NumberOfDonations FROM DONOR D JOIN blood_bags BB ON D.D_CNIC = BB.D_CNIC WHERE D.Blood_Group = 'A+' GROUP BY D.Name, D.Contact, D.Blood_Group ORDER BY NumberOfDonations DESC;</pre>
Output	<pre>+-----+-----+-----+-----+ Name Contact Blood_Group NumberOfDonations +-----+-----+-----+-----+ M Saleem 0301-2685998 A+ 1 muhammad Usman 0301-7720324 A+ 1 Huma Saleem 0345-6788431 A+ 1 +-----+-----+-----+-----+ 3 rows in set (0.00 sec)</pre>
3	Find all blood bags delivered on a specific date (e.g., '2024-06-05').
Query	<pre>SELECT Bag_ID, Donation_Date, Delivery_Date FROM BLOOD_BAGS WHERE Delivery_Date = '2024-06-05';</pre>
Output	<pre>+-----+-----+-----+ Bag_ID Donation_Date Delivery_Date +-----+-----+-----+ 12 2024-06-02 2024-06-05 +-----+-----+-----+ 1 row in set (0.00 sec)</pre>
4	Count the number of blood bags donated by each donor.
Query	<pre>SELECT d.Name AS 'Donor Name', COUNT(bb.Bag_ID) AS Blood_Bag_Count FROM DONOR d JOIN BLOOD_BAGS bb ON d.D_CNIC = bb.D_CNIC GROUP BY d.Name;</pre>

Output	<pre> +-----+-----+ Donor Name Blood_Bag_Count +-----+-----+ M Saleem 1 muhammad Usman 1 riaz u din 1 Huma Saleem 1 Aqsa aziz 1 Ghazia 1 Samra Tariq 1 Shahida 1 Bushra 1 +-----+-----+ 9 rows in set (0.00 sec) </pre>
5	Find the number of blood bags managed by each staff member.
Query	<pre> SELECT s.S_ID AS 'Staff_ID', s.Name, COUNT(bb.Bag_ID) AS Blood_Bag_Manage FROM STAFF s JOIN BLOOD_BAGS bb ON s.S_ID = bb.S_ID GROUP BY s.S_ID, s.Name; </pre>
Output	<pre> +-----+-----+-----+ Staff_ID Name Blood_Bag_Manage +-----+-----+-----+ 1 Shazam 1 2 Abdul Raheem 3 3 Rehan 2 4 Awais 2 5 Ghazia 2 6 Ali Taimoor 2 7 Anila 1 8 Nouman 1 9 Shariq Khan 1 +-----+-----+-----+ 9 rows in set (0.00 sec) </pre>
6	Find all blood bags provided by a specific network (e.g., network ID 3).
Query	<pre> SELECT COUNT(bb.Bag_ID) AS Total_Bags, n.N_ID AS Network_ID, n. Institute_Name, bb.Donation_Date FROM BLOOD_BAGS bb JOIN NETWORKS n ON bb.Bag_ID = n.N_ID WHERE n.N_ID = 3; </pre>
Output	<pre> +-----+-----+-----+-----+ Total_Bags Network_ID Institute_Name Donation_Date +-----+-----+-----+-----+ 1 3 Red Crescent Blood Bank Mianwali 2024-05-23 +-----+-----+-----+-----+ 1 row in set (0.01 sec) </pre>
7	List all blood bags ordered by their expiry date.

Query	SELECT bb.Bag_ID, bb.Donation_Date, DATE_ADD(bb.Donation_Date, INTERVAL 42 DAY) AS Expiry_Date FROM BLOOD_BAGS bb ORDER BY Expiry_Date;
Output	<pre> +-----+-----+-----+ Bag_ID Donation_Date Expiry_Date +-----+-----+-----+ 1 2024-05-20 2024-07-01 2 2024-05-21 2024-07-02 3 2024-05-23 2024-07-04 4 2024-05-24 2024-07-05 5 2024-05-24 2024-07-05 6 2024-05-27 2024-07-08 7 2024-05-29 2024-07-10 8 2024-05-31 2024-07-12 9 2024-06-01 2024-07-13 10 2024-06-01 2024-07-13 11 2024-06-01 2024-07-13 12 2024-06-02 2024-07-14 13 2024-06-03 2024-07-15 14 2024-06-03 2024-07-15 15 2024-06-04 2024-07-16 +-----+-----+-----+ 15 rows in set (0.00 sec) </pre>
8	Query to Find Eligible Donors
Query	SELECT d.D_CNIC AS Donor_identity, d.Name AS Donor_Name, Contact, d.Blood_Group, 'Eligible' AS Eligibility_Status FROM DONOR d WHERE NOT EXISTS (SELECT 1 FROM BLOOD_BAGS bb WHERE bb.D_CNIC = d.D_CNIC) OR (SELECT MAX(bb.Donation_Date) FROM BLOOD_BAGS bb WHERE bb.D_CNIC = d.D_CNIC) <= DATE_SUB(CURRENT_DATE, INTERVAL 90 DAY);
Output	<pre> +-----+-----+-----+-----+-----+ Donor_identity Donor_Name Contact Blood_Group Eligibility_Status +-----+-----+-----+-----+-----+ 30014-0000014-1 M Irfan 0335-6302675 A+ Eligible 30018-0000018-5 Muhammad Usman 0345-6443752 O+ Eligible 30047-0000047-2 Aqsa 0311-7121603 A+ Eligible 30049-0000049-4 sana Malik 0324-5972465 A+ Eligible 30069-0000069-6 Ahad Khan 0309-6556446 A+ Eligible 30076-0000076-2 Khan Mehmood 0307-6566654 A+ Eligible 30092-0000092-2 Sana 0349-6084322 A+ Eligible 30104-0000104-5 Siuaira 0301-5840522 A+ Eligible 30164-0000164-2 Quratulain 0306-8454860 AB- Eligible 30167-0000167-5 Nisar Ali 0345-4779109 B+ Eligible 30214-0000214-6 Naseem 0347-7611603 O+ Eligible 30220-0000220-4 Bisma amir 0323-1897256 B+ Eligible 30225-0000225-9 bilal hassan 0349-9669755 B+ Eligible 30244-0000244-5 Muhammad Ismail 0324-6457297 B+ Eligible 30258-0000258-6 Naila 0345-2017983 B+ Eligible +-----+-----+-----+-----+-----+ </pre>

	<pre> 30534-0000534-0 Nida Anwar 0308-7981364 A+ Eligible 30535-0000535-1 Shahbaz 0343-7743305 B+ Eligible 30543-0000543-4 Salma 0305-6291897 O+ Eligible 30547-0000547-3 Naeem 0342-9574186 A+ Eligible 30550-0000550-5 Aziz Ul Haq 0321-9451601 O+ Eligible 30562-0000562-4 Wazir M 0333-7363017 O+ Eligible 30568-0000568-1 basit 0327-0367753 O+ Eligible 30569-0000569-2 Aqsa 0327-3299072 O+ Eligible 30573-0000573-6 Rehana 0325-8174066 O+ Eligible +-----+-----+-----+-----+ 43 rows in set (0.00 sec) </pre>
9	Find the total number of blood bags donated each month.
Query	<pre> SELECT DATE_FORMAT(bb.Donation_Date, '%Y-%m') AS Month, COUNT(bb.Bag_ID) AS Blood_Bag_Count FROM BLOOD_BAGS bb GROUP BY DATE_FORMAT(bb.Donation_Date, '%Y-%m'); </pre>
Output	<pre> +-----+-----+ Month Blood_Bag_Count +-----+-----+ 2024-05 8 2024-06 7 +-----+-----+ 2 rows in set (0.01 sec) </pre>
10	This query retrieves blood bag information along with the donor's name and blood group, calculates the expiry date as 30 days after the delivery date, and counts the number of bags for each unique bag ID, then orders the results by the expiry date.
Query	<pre> SELECT BB.Bag_ID, BB.Delivery_Date, ADDEDTE(BB.Delivery_Date, 30) AS Expiry_date, D.Name AS Donor_Name, D.Blood_Group, COUNT(BB.Bag_ID) AS NumberOfBags FROM blood_bags BB JOIN DONOR D ON BB.D_CNIC = D.D_CNIC GROUP BY BB.Bag_ID, BB.Delivery_Date, D.Name, D.Blood_Group ORDER BY Expiry_date; </pre>

Output	<pre> +-----+-----+-----+-----+-----+-----+ Bag_ID Delivery_Date Expiry_date Donor_Name Blood_Group NumberOfBags +-----+-----+-----+-----+-----+-----+ 3 NULL NULL riaz u din AB+ 1 8 NULL NULL Samra Tariq B- 1 9 NULL NULL Huma Saleem A+ 1 2 2024-05-22 2024-06-21 muhammad Usman A+ 1 1 2024-05-30 2024-06-29 M Saleem A+ 1 5 2024-06-01 2024-07-01 Aqsa aziz B+ 1 7 2024-06-03 2024-07-03 Ghazia B+ 1 12 2024-06-05 2024-07-05 Shahida B+ 1 14 2024-06-12 2024-07-12 Bushra B+ 1 +-----+-----+-----+-----+-----+-----+ 9 rows in set (0.00 sec) </pre>
11	Query to return detail of those staff member who have collected blood from networks.
Query	SELECT S.S_ID, S.NAME, S.CONTACT FROM STAFF S WHERE S.S_ID IN(SELECT B.S_ID FROM BLOOD_BAGS B WHERE B.N_ID IS NOT NULL);
Output	<pre> +-----+-----+-----+ S_Id name contact +-----+-----+-----+ 9 Shariq Khan 0321-1234575 7 Anila 0321-1234573 4 Awais 0321-1234570 8 Nouman 0321-1234574 6 Ali Taimoor 0321-1234572 +-----+-----+-----+ 5 rows in set (0.01 sec) </pre>
12	Give detail of donor and recipient having delivery date not null.
Query	SELECT D.D_CNIC, D.BLOOD_GROUP, R.R_CNIC,R.NAME, R.INSTITUTE FROM DONOR D, RECIPIENT R WHERE (D.D_CNIC, R.R_CNIC) IN (SELECT D_CNIC, R_CNIC FROM BLOOD_BAGS WHERE DELIVERY_DATE IS NOT NULL);
Output	<pre> +-----+-----+-----+-----+-----+ D_CNIC blood_group R_CNIC name institute +-----+-----+-----+-----+-----+ 30005-0000005-5 A+ 33848-1234567-0 Madiha Tasneem hospital, bun 30033-0000033-6 A+ 34929-7654321-1 Sami Tabish Children Hospital, Mianwali 30316-0000316-1 B+ 37515-3698741-7 Maryam Sarmad Trust Blood Bank 30336-0000336-3 B+ 32737-7654321-9 Lubna Tabish Children Hospital, Mianwali 30507-0000507-2 B+ 33838-2468013-0 Eman Rehman Hospital, Mianwali 30553-0000553-9 B+ 36505-7890123-7 Zainab Tasneem hospital, bun +-----+-----+-----+-----+-----+ 6 rows in set (0.00 sec) </pre>
13	Query to select details of those recipient who have received blood group 'A+'.
Query	SELECT * FROM RECIPIENT WHERE R_CNIC IN (SELECT R_CNIC FROM

	BLOOD_BAGS WHERE D_CNIC IN (SELECT D_CNIC FROM DONOR WHERE BLOOD_GROUP='A+'));
Output	<pre> +-----+-----+-----+-----+-----+ R_CNIC NAME CONTACT ADDRESS INSTITUTE +-----+-----+-----+-----+-----+ 33848-1234567-0 Madiha 0368-5678901 Ballo khail RD, Minawali Tasneem hospital, bun 34929-7654321-1 Sami 0349-1098765 Gulberg Chock, Minawali Tabish Children Hospital, Mianwali +-----+-----+-----+-----+-----+ 2 rows in set (0.00 sec) </pre>
14	This query is using INNER JOIN to find those blood bag IDs that are donated by institutes with institute details.
Query	SELECT N.N_ID, N.INSTITUTE_NAME, N.CONTACT, B.BAG_ID FROM NETWORKS N INNER JOIN BLOOD_BAGS B ON N.N_ID=B.N_ID;
Output	<pre> +-----+-----+-----+-----+ N_id institute_name contact bag_id +-----+-----+-----+-----+ 1 Blood Bank DHQ Hospital Mianwali (0459) 235136 10 1 Blood Bank DHQ Hospital Mianwali (0459) 235136 11 2 Naml Wellness Blood Bank (0459) 229847 15 3 Red Crescent Blood Bank Mianwali (0459) 228035 4 4 Sarmad Trust Blood Bank (0459) 226043 13 5 Agha Khan Blood Bank Mianwali (0459) 227589 6 +-----+-----+-----+-----+ 6 rows in set (0.00 sec) </pre>

CHAPTER 5 : INTERFACE DESIGN

5.1. LANGUAGE/Framework:

For the development of our GUI client, we chose the Python programming language with the Tkinter library for the graphical user interface. Python was selected due to its simplicity, readability, and vast ecosystem of libraries, which accelerates development and enhances productivity. Tkinter, being the standard GUI toolkit for Python, offers a straightforward and efficient way to create graphical interfaces.

Suitability and Advantages:

- **Ease of Use:** Python's syntax is clear and concise, making it easier for developers to write and maintain the code.
- **Extensive Libraries:** Python offers a wide range of libraries for database connectivity, GUI development, and other functionalities.
- **Cross-Platform Compatibility:** Both Python and Tkinter are cross-platform, ensuring that the application can run on various operating systems without significant changes.
- **Community Support:** Python has a large community, providing plenty of resources, tutorials, and third-party modules.

Key Features:

- **Event-Driven Programming:** Tkinter supports event-driven programming, allowing the application to respond to user actions seamlessly.
- **Widgets:** Tkinter offers a variety of widgets (e.g., buttons, labels, text boxes) that are essential for building a user-friendly interface.
- **Customization:** Tkinter allows for the customization of widgets and layouts, providing the flexibility to design an intuitive and attractive GUI.

These features contribute significantly to the development and functionality of the GUI client by enabling us to create an interactive and responsive interface for both admin and user sides of the application.

5.2. DATABASE CONNECTIVITY:

The GUI client connects to the MySQL database using the mysql-connector-python library. This library provides a robust interface for connecting to MySQL databases from Python, handling various database operations, and ensuring secure and efficient data transactions.

Code Snippet for Database Connection:

```
import mysql.connector  
  
from mysql.connector import Error
```

```
def create_connection():
    connection = None
    try:
        connection = mysql.connector.connect(
            host="localhost",
            user="root",
            password="Namal.123",
            database="namal_blood_bank"
        )
        if connection.is_connected():
            print("Connection to MySQL DB successful")
    except Error as e:
        print(f"The error '{e}' occurred")
    return connection

# Example usage
conn = create_connection()
```

Error Handling:

The above code snippet includes basic error handling by catching any `Error` exceptions that occur during the connection attempt. This ensures that any issues with the database connection are identified and reported, allowing for troubleshooting.

5.3. STORED PROCEDURES AND FUNCTIONS:

5.3.1. Stored Procedure: GetEligibleDonors

Purpose:

This procedure retrieves donors who are eligible based on specific criteria (either no donations recorded or last donation date more than 90 days ago).

Script:

DELIMITER //

```
CREATE PROCEDURE GetEligibleDonors()
BEGIN
    SELECT
        d.D_CNIC AS Donor_identity,
        d.Name AS Donor_Name,
        d.Contact,
        d.Blood_Group,
        'Eligible' AS Eligibility_Status
    FROM
        DONOR d
    WHERE
        NOT EXISTS (
            SELECT 1
            FROM BLOOD_BAGS bb
            WHERE bb.D_CNIC = d.D_CNIC
        )
        OR (
            SELECT MAX(bb.Donation_Date)
            FROM BLOOD_BAGS bb
            WHERE bb.D_CNIC = d.D_CNIC
        ) <= DATE_SUB(CURRENT_DATE, INTERVAL 90 DAY);
END //
```

DELIMITER ;

Implementation:

- **Objective:** This procedure identifies donors who can potentially donate blood based on their donation history.
- **How it Enhances Functionality:** It allows the GUI client to easily fetch and display eligible donors, ensuring efficient management of blood availability.

5.3.2. Stored Procedure: GetBloodBagCountsByMonth

Purpose:

This procedure aggregates and counts blood bags donated each month.

Script:

```
DELIMITER //
```

```
CREATE PROCEDURE GetBloodBagCountsByMonth()
BEGIN
    SELECT
        DATE_FORMAT(bb.Donation_Date, '%Y-%m') AS Month,
        COUNT(bb.Bag_ID) AS Blood_Bag_Count
    FROM
        BLOOD_BAGS bb
    GROUP BY
        DATE_FORMAT(bb.Donation_Date, '%Y-%m');
END //
```

```
DELIMITER ;
```

Implementation:

- **Objective:** To provide a monthly summary of blood bag donations.
- **How it Enhances Functionality:** It enables the GUI client to visualize and track blood donation trends over time, aiding in inventory management and planning.

5.3.3. Stored Procedure: GetBloodBagDetails

Purpose:

This procedure retrieves detailed information about blood bags, including donor information.

Script:

```
DELIMITER //
```

```
CREATE PROCEDURE GetBloodBagDetails()
BEGIN
    SELECT
        BB.Bag_ID,
        BB.Delivery_Date,
        ADDDATE(BB.Delivery_Date, 30) AS Expiry_date,
        D.Name AS Donor_Name,
        D.Blood_Group,
        COUNT(BB.Bag_ID) AS NumberOfBags
    FROM
        BLOOD_BAGS BB
    JOIN
        DONOR D ON BB.D_CNIC = D.D_CNIC
    GROUP BY
        BB.Bag_ID, BB.Delivery_Date, D.Name, D.Blood_Group
    ORDER BY
```

```
    Expiry_date;  
END //
```

DELIMITER ;

Implementation:

- **Objective:** To retrieve comprehensive details of blood bags, including donor names, blood groups, delivery dates, and expiry dates.
- **How it Enhances Functionality:** It provides the GUI client with essential information for managing blood stock efficiently, facilitating quick access to critical data.

5.3.4. Function: CalculateExpiryDate

Purpose:

This function calculates the expiry date of a blood bag based on its delivery date.

Script:

```
CREATE FUNCTION CalculateExpiryDate(delivery_date DATE)  
RETURNS DATE  
BEGIN  
    DECLARE expiry_date DATE;  
    SET expiry_date = ADDDATE(delivery_date, 30);  
    RETURN expiry_date;  
END;
```

Implementation:

- **Objective:** To compute the expiry date of a blood bag, typically 30 days from its delivery date.
- **How it Enhances Functionality:** It can be used within SQL queries or other procedures/functions to dynamically calculate and manage blood bag expiry dates, ensuring accuracy in blood stock management.

Explanation:

- **Stored Procedures:** These encapsulate complex queries or operations that are frequently used in the application, promoting code reusability and reducing network overhead by executing the logic directly on the database server.
- **Functions:** Provide reusable calculations or transformations that can be incorporated into SQL queries or procedures, enhancing flexibility and maintainability of the database logic.

The database utilizes stored procedures and functions to enhance the application's functionality by deriving expiry blood bags and identifying eligible donors.

Objective and Implementation In GUI:

- **GetExpiredBags Method:** This procedure retrieves all blood bags that have an expiry date earlier than the current date, helping the admin to identify and remove expired blood bags from the inventory.
- **EligibleDonors Method:** This function returns a list of donors who are eligible to donate blood based on their last donation date, ensuring that only suitable donors are contacted.

These scripts are implemented in the MySQL database and are called from the Python application as needed to enhance the GUI client's functionality.

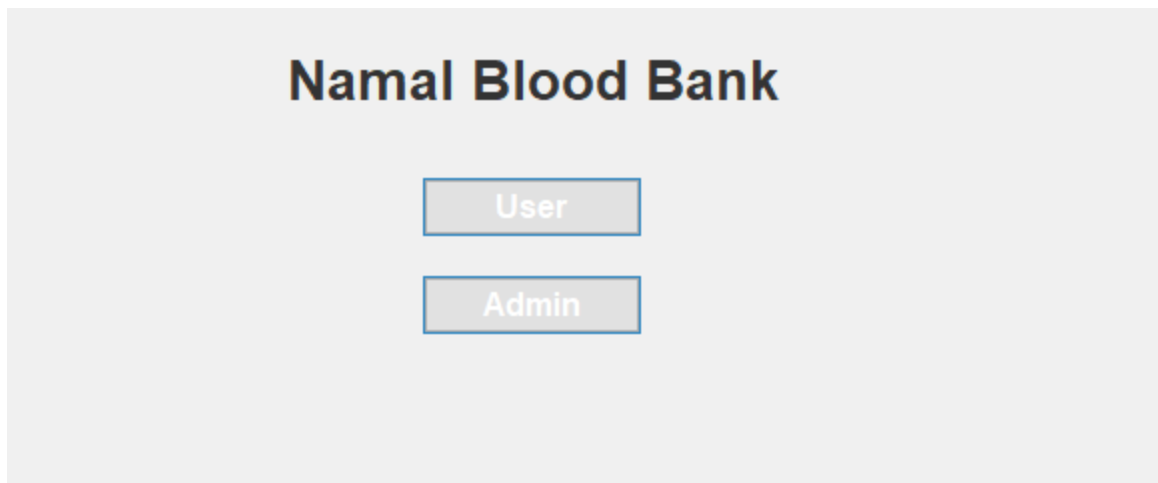
5.4. INTERFACES: [3]

Below are screenshots and descriptions of the main interfaces of our GUI client.

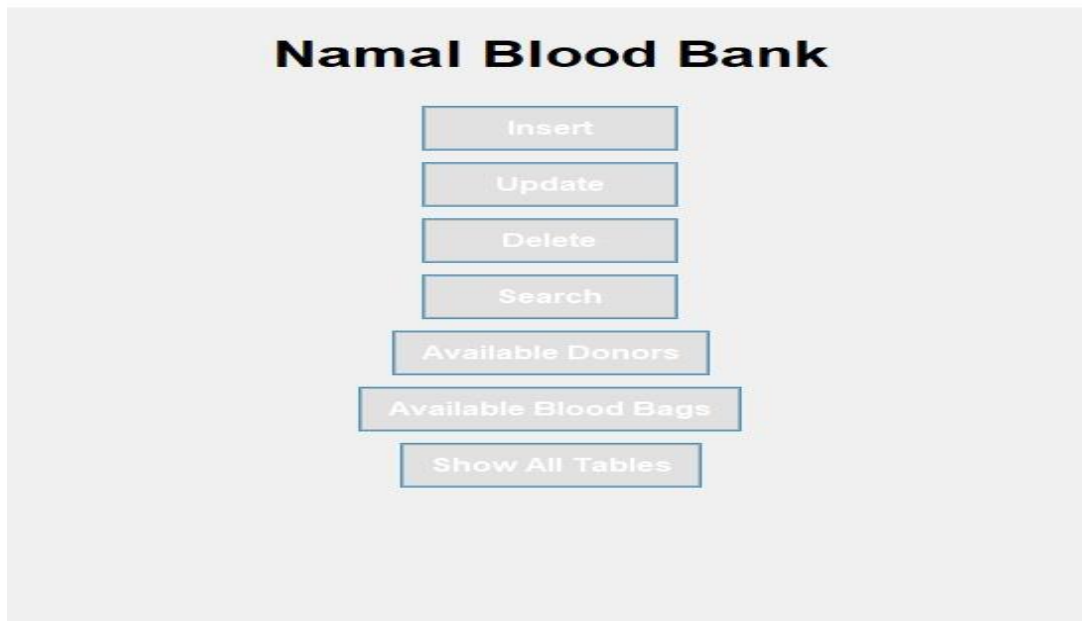
Main Interface:

1) Admin Interface

2) User Interface



5.4.1. Admin Interface



1. Forms for Data Entry

- **Description:** This screenshot shows the forms used for entering new data into the database. Each form includes fields for relevant attributes such as donor information, blood bag details, recipient data, etc. Data validation checks ensure accurate input.

The screenshot shows three panels of the Namal Blood Bank interface. The left panel, titled 'Namal Blood Bank', contains a vertical list of buttons: New Donor, New Recipient, New Network, New Staff, and New Blood Bag. The middle panel, titled 'NEW DONOR', contains a form with the following fields: CNIC, Name, Contact, Department, Category, and Blood Group. Below the form is an 'Add Donor' button. The right panel, titled 'NEW STAFF', contains a form with the following fields: Name, Contact, CNIC, and Address. Below the form is an 'Add Staff' button.

The image displays three separate forms for adding new data to the system, arranged side-by-side on a light gray background. Each form has a title in green text at the top.

- NEW NETWORK:** Contains three input fields labeled 'Institute Name', 'Contact', and 'Address'. Below the form is a label 'Add Network' and a green button labeled 'Add Network'.
- NEW BLOOD BAG:** Contains five input fields labeled 'Donation Date (YYYY-MM-DD)', 'Delivery Date (YYYY-MM-DD)', 'Staff ID', 'Donor CNIC', and 'Recipient CNIC'. Below the form is a label 'Add Blood Bag' and a green button labeled 'Add Blood Bag'.
- NEW RECIPIENT:** Contains four input fields labeled 'CNIC', 'Name', 'Contact', and 'Address'. Below the form is a label 'Add Recipient' and a green button labeled 'Add Recipient'.

2. Display Existing Data

- **Description:** Here, existing data from the database is displayed in a table format. This view allows administrators to see all records currently stored, such as blood bag records, donor information, staff details, etc.

The image shows a main menu for displaying existing data. It features a large title 'Namal Blood Bank' at the top. Below the title, there are five buttons stacked vertically, each with a light blue background and a dark blue border. The buttons are labeled: 'Show All Donors', 'Show All Recipients', 'Show All Networks', 'Show All Staff', and 'Show All Blood Bags'.

Namal Blood Bank Management System (NBBMS)

D_CNIC	Name	Contact	Department	Category	Blood_Group
30001-000001-1	Adnan Iqbal	09877234567			
30002-000002-2	Muhammad Saleem	0301-2685998		Staff	A+
30003-000003-3	Muhammad Saleem	0301-2685998		Staff	A+
30004-000004-4	Yaar Muhammad Khan	0301-7698303		Staff	A+
30005-000005-5	M Saleem	0301-2685998		Faculty	A+
30006-000006-6	Naveed Ahmad	0345-9665509		Staff	A+
30007-000007-7	said Anwar	0302-6065103	CS	4th Year	A+
30008-000008-8	Muhammad Awais	0307-9062748	Math	4th Year	A+
30009-000009-9	Hanif Noor	0332-0830144	CS	4th Year	A+
30010-000010-1	Muhammad Dawood sewood	0310-6760423	CS	4th Year	A+
30011-000011-2	Mishkatullah	0345-9529168	Maths	4th Year	A+
30012-000012-3	naila saif	0312-5535565	maths	4th Year	A+
30013-000013-4	muhammad zain	0308-5363655	math	4th Year	A+
30014-000014-5	javeria amir	0324-2435363	math	4th Year	A+
30015-000015-6	amjad iqbal	0303-0574811	math	4th Year	A+
30016-000016-7	muhammad shafqat	0316-1613442	EE	4th Year	A+
30017-000017-8	wasat ullah	0340-2692916	BBA	4th Year	A+
30018-000018-9	hareem naz	0312-4646435	BBA	4th Year	A+
30020-000020-2	Bakhat Maarj	0344-4346353	Maths	3rd Year	A+
30021-000021-3	Nargis Shehzadi	0324-2281565	Maths	3rd Year	A+
30022-000022-4	M. Bilal khan	0308-2684617	Maths	3rd Year	A+
30023-000023-5	Mohsin Zareef Khan	0341-6756671	Maths	3rd Year	A+
30024-000024-6	Bushra Khadim	0304-4971200	Maths	3rd Year	A+
30025-000025-7	Mehwish Iqbal	0321-6696360	Maths	3rd Year	A+
30026-000026-8	M. Tufail Khan	0312-4345310	Maths	3rd Year	A+

N_ID	Institute_Name	Contact	Address
1	Blood Bank DHQ Hospital Mianwali	(0459) 235136	HQJV+639, Mianwali, Punjab
2	NamI Wellness Blood Bank	(0459) 229847	NamI College Road, Mianwali
3	Red Crescent Blood Bank Mianwali	(0459) 228035	Main Street, Mianwali
5	AS1	3232	SADA

Bag_ID	Donation_Date	Delivery_Date	S_ID	D_CNIC	R_CNIC	N_ID
2	2024-05-21	2024-05-22	2	30346-0000346-4	36131-3698741-3	None
3	2024-05-23	None	3	30049-0000049-4	None	None
4	2024-05-24	2024-06-03	4	30069-0000069-6	38424-1234567-6	None
6	2024-05-27	None	6	None	None	5
7	2024-05-29	2024-06-03	5	30134-0000134-8	34020-1357902-2	None
8	2024-05-31	None	3	30295-0000295-7	None	None
9	2024-06-01	None	2	30033-0000033-6	None	None
10	2024-06-01	2024-06-22	9	None	35080-1234567-2	1
11	2024-06-01	2024-06-20	7	30220-0000220-4	35080-1234567-2	1
12	2024-06-02	2024-06-05	2	30092-0000092-2	35080-1234567-2	1
13	2024-06-03	2024-06-08	8	30568-0000568-1	35080-1234567-2	1
14	2024-06-03	2024-06-12	6	30347-0000347-5	35080-1234567-2	1
15	2024-06-04	2024-06-10	4	30121-0000121-4	35080-1234567-2	1
16	2024-06-06	None	3	30068-0000068-5	None	None
17	2024-06-07	2024-06-15	5	30422-0000422-8	35080-1234567-2	1
18	2024-06-08	None	1	30330-0000330-6	None	None
19	2024-06-09	2024-06-20	9	30379-0000379-1	35080-1234567-2	1
20	2024-06-09	2024-06-25	7	30227-0000227-2	35080-1234567-2	1
21	2024-06-10	2024-06-18	2	30203-0000203-5	35080-1234567-2	1

S_ID	NAME	CONTACT	CNIC	ADDRESS
1	Nahan	0301-1234567	32101-7654321-4	Namal University Mianwali
2	Abdul	0301-1234567	32101-7654321-5	Namal University Mianwali
5	Ghazia	0321-1234571	32101-7654321-6	Namal University Mianwali
6	Ali Taimoor	0321-1234572	32101-7654321-7	Namal University Mianwali
7	Amelia	0334-1234573	33404-7664321-8	Namal University Mianwali
8	Nauman	0321-1234574	32101-7654321-9	Namal University Mianwali
9	Shiruk Khan	0321-1234575	32101-7654321-1	Namal University Mianwali
10	Umar Shehzad	0321-1234576	32101-7654321-1	Namal University Mianwali
11	Zoya	0321-1234577	32101-7654321-6	Namal University Mianwali
12	Shahid	0321-1234578	32101-7654321-1	Namal University Mianwali
13	Umair Ahmed	0321-1234579	32101-7676321-4	Namal University Mianwali
14	Kamran Malik	0321-1234580	32101-7657321-5	Namal University Mianwali
15	Asim Khan	0321-1234581	32101-7666321-8	Namal University Mianwali
16	Noor Fatima	0321-1234582	32101-7654321-6	Namal University Mianwali
18	Asim Khan	03178818644		Namal
19	Shahid	0321-1234578	32101-7654321-1	Namal University Mianwali
20	Shahid	0321-1234578	32101-7654321-1	Namal University Mianwali
21	Shahid	0321-1234578	32101-7654321-1	Namal University Mianwali
22	Shahid	0321-1234578	32101-7654321-1	Namal University Mianwali
23	Shahid	0321-1234578	32101-7654321-1	Namal University Mianwali
24	Shahid	0321-1234578	32101-7654321-1	Namal University Mianwali
25	Shahid	0321-1234578	32101-7654321-1	Namal University Mianwali
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99	Shahid	0321-1234578	32101-7654321-1	Namal University Mianwali
100	Shahid	0321-1234578	32101-7654321-1	Namal University Mianwali

Namal Blood Bank Management System (NBBMS)

Recipient Table

R_CNIC	NAME	CONTACT	ADDRESS	INSTITUTE
12345	umair	34577	bhakkar	
30535-3698521-7	Noor	0355-8901234	Dhumaka, Minawali	Kashaf Homoeopathic Clinic, Mianwali.
30545-7890123-7	Hira	0365-3456789	Agha khan center, Minawali	Agha Khan Blozod Bank Mianwali
30555-3698741-7	Laila	0375-8901234	HGJV+639, Mianwali	Blood Bank DHQ Hospital Mianwali
30565-7654321-7	Nadia	0385-3456789	Dhumaka, Minawali	Kashaf Homoeopathic Clinic, Mianwali
30575-3698521-7	Laila	0395-8901234	Agha khan center, Minawali	Agha Khan Blozod Bank Mianwali
30585-7890123-7	Hira	0405-3456789	HGJV+639, Mianwali	Blood Bank DHQ Hospital Mianwali
30595-3698741-7	Nadia	0415-8901234	Dhumaka, Minawali	Kashaf Homoeopathic Clinic, Mianwali
31626-2468013-8	Javed	0346-9876543	HGJV+639, Mianwali	Blood Bank DHQ Hospital Mianwali
31636-1357902-8	Yasir	0356-4321098	Gulberg Chock, Minawali	Rehman Hospital, Mianwali
31646-2468135-8	Khalid	0366-9876543	Gulberg Chock, Minawali	Obaid Noor Hospital, Mianwali
31656-1234567-8	Amir	0376-4321098	HGJV+639, Mianwali	Blood Bank DHQ Hospital Mianwali
31666-2468013-8	Salman	0386-9876543	Gulberg Chock, Minawali	Rehman Hospital, Mianwali
31676-1357902-8	Waqar	0396-4321098	Gulberg Chock, Minawali	Obaid Noor Hospital, Mianwali
31686-2468135-8	Umar	0406-9876543	HGJV+639, Mianwali	Blood Bank DHQ Hospital Mianwali
31696-1234567-8	Salman	0416-4321098	Gulberg Chock, Minawali	Rehman Hospital, Mianwali
31818-2468013-0	Umar	0338-5432109	Dhumaka, Minawali	Kashaf Homoeopathic Clinic, Mianwali
31909-7654321-1	Saira	0329-5678901	Gulberg Chock, Minawali	Rehman Hospital, Mianwali
32010-2468013-2	Arman	0330-1098765	Gulberg Chock, Minawali	Obaid Noor Hospital, Mianwali
32101-7654321-2	Adnan	0321-1234567	Ballo khail RD, Minawali	Tabish Children Hospital, Mianwali
32727-3698521-9	Alina	0347-4567890	Ballo khail RD, Minawali	Tabish Children Hospital, Mianwali
32737-7890123-9	Farhana	0357-9012345	Namal, Minawali	Namal hospital
32747-3698741-9	Saba	0367-4567890	Gulberg Chock, Minawali	Rehman Hospital, Mianwali
32757-7654321-9	Saira	0377-9012345	Ballo khail RD, Minawali	Tabish Children Hospital, Mianwali
32767-3698521-9	Saba	0387-4567890	Namal, Minawali	Namal hospital
32777-7890123-9	Sadia	0397-9012345	Gulberg Chock, Minawali	Rehman Hospital, Mianwali

3. Search Data

- Description:** The search feature enables administrators to search for specific records based on various criteria. It enhances usability by quickly locating relevant information without manually scanning through large datasets.

Namal Blood Bank

Search Donor

Search Recipient

Search Network

Search Staff

Search Blood Bag

Namal Blood Bank Management System (NBBMS)

Network Records

Enter N_ID:

N_ID	Institute_Name	Contact	Address
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Blood Bag Records

Enter Bag_ID:

Bag_ID	Donation_Date	Delivery_Date	S_ID	D_CNIC	R_CNIC	N_ID
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Enter D_CNIC:

Namal Blood Bank Management System (NBBMS)

The top screenshot displays the 'Recipient Records' window. It features a search bar labeled 'Enter R_CNIC:' with a green 'Search' button. Below the search bar is a table with the following columns: R_CNIC, NAME, CONTACT, ADDRESS, and INSTITUTE. The table is currently empty.

The bottom screenshot displays the 'Staff Records' window. It features a search bar labeled 'Enter S_ID:' with a green 'Search' button. Below the search bar is a table with the following columns: S_ID, NAME, CONTACT, CNIC, and ADDRESS. The table is currently empty.

4. Modify Data

- **Description:** This screenshot illustrates the functionality to modify existing records. Admins can select a record from the displayed data, edit its details directly within the form fields, and save the changes back to the database.

The screenshot shows the 'Namal Blood Bank' management interface. It features a title 'Namal Blood Bank' at the top. Below the title are five buttons arranged vertically: 'Update Donor', 'Update Recipient', 'Update Network', 'Update Staff', and 'Update Blood Bag'.

The screenshot displays the 'Update Network' page of the NBBMS. It features three distinct forms for updating different components of the blood bank network:

- UPDATE STAFF INFORMATION:** Includes fields for Staff ID, Name, Contact, CNIC, and Address. A green 'Update Staff' button is located below the form.
- UPDATE RECIPIENT INFORMATION:** Includes fields for CNIC, Name, Contact, Address, and Institute. A green 'Update Recipient' button is located below the form.
- NETWORK DETAILS:** Includes fields for Network ID, Institute Name, Contact, and Address. A green 'Update Network' button is located below the form.

Each form is contained within a white box with a green title, and the buttons are green with white text. The background is a light gray.

5. Delete Data

- **Description:** Admins can delete records from the database using this feature. By selecting a record and confirming deletion, the system removes the entry from the database, ensuring data integrity and management.

Namal Blood Bank

Delete Donor

Delete Network

Delete Staff

DELETE STAFF
Staff ID

Delete Staff

Delete Staff

REMOVE DONOR
Donor CNIC

Delete Donor

Delete Donor

I Bank - Delete Network

REMOVE NETWORK
Network ID

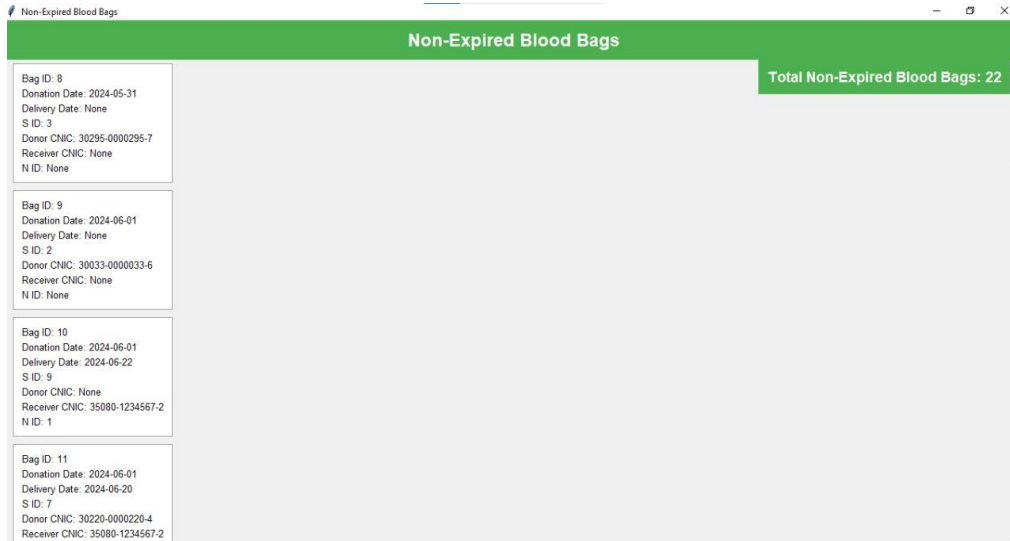
Delete Network

Delete Network

5.4.2. User Interface

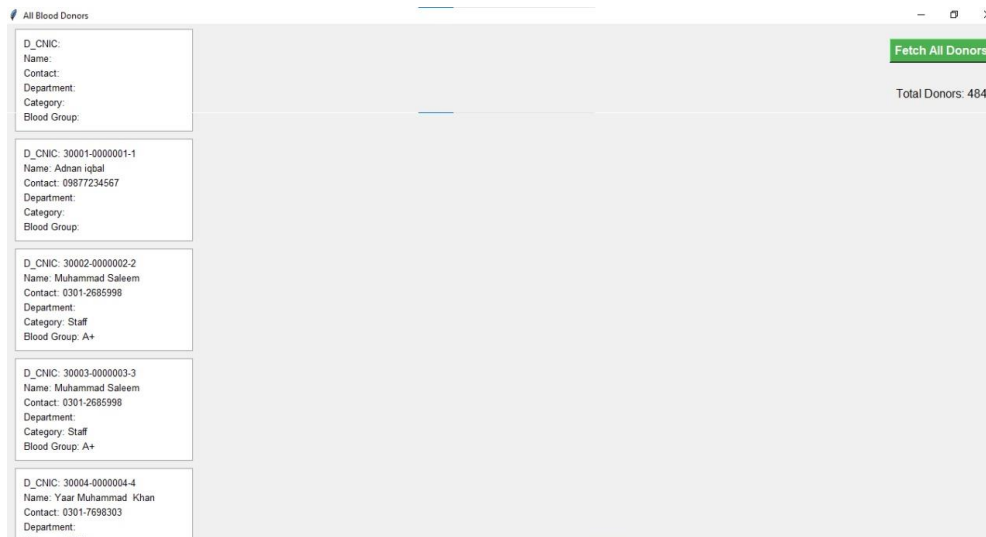
1. View Available Blood Bags

- **Description:** The user interface allows users to view available blood bags. It displays essential details such as blood type, and expiration date, enabling users to find and select suitable blood bags for recipients.



2. Search for Donors via Blood Bags

- **Description:** Users can search for donors through blood bags. This functionality helps users find donors based on specific blood types or other criteria associated with blood bags in the system.



CHAPTER 6 : CONCLUSION

6.1. LESSONS LEARNED:

Throughout the development of the Namal Blood Bank Management System (NBBMS), several key lessons were gleaned that encompass technical insights, project management skills, teamwork experiences, and overall project execution.

Firstly, from a technical perspective, the project reinforced the importance of robust database design and efficient query management. Understanding how to optimize database operations significantly enhanced the system's performance and responsiveness.

In terms of project management, effective communication and regular progress tracking emerged as crucial factors in ensuring team cohesion and project success. Clear delegation of tasks among team members, facilitated by regular meetings and updates, helped in maintaining project timelines and addressing issues promptly.

Teamwork experiences underscored the significance of collaboration and leveraging individual strengths. Assigning roles based on expertise allowed each team member to contribute effectively, fostering a productive work environment.

Lastly, the project highlighted the value of adaptability and problem-solving skills. Adapting to unexpected challenges during implementation and testing phases required innovative solutions and the ability to pivot when necessary.

6.2. CHALLENGES AND SOLUTIONS:

Throughout the project lifecycle, several challenges tested the team's resolve and ingenuity.

During the design phase, defining the database schema and ensuring compatibility with all required functionalities posed initial challenges. Through meticulous planning and consultation with the database designer, these challenges were overcome by iteratively refining the schema to accommodate evolving project requirements.

Implementation challenges included integrating diverse subsystems and ensuring seamless interaction between the frontend and backend components. Issues such as data consistency and user interface responsiveness were resolved through rigorous testing and debugging sessions. Collaborative troubleshooting and peer reviews played a pivotal role in identifying and rectifying implementation errors promptly.

Testing presented its own set of challenges, particularly in simulating real-world scenarios and edge cases. Rigorous testing protocols and the adoption of automated testing frameworks helped in identifying and addressing bugs early in the development cycle. Additionally, user feedback sessions provided valuable insights for refining user interface elements and enhancing user experience.

6.3. FUTURE WORK AND IMPROVEMENTS:

There are several avenues for enhancing the NBBMS to better serve its users and stakeholders.

One area of improvement involves enhancing data analytics capabilities within the system. Implementing data visualization tools and predictive analytics algorithms could provide deeper insights into blood donation patterns and inventory management, thereby optimizing resource allocation.

Furthermore, integrating mobile-friendly interfaces and developing a dedicated mobile application could enhance accessibility and user engagement, catering to a broader audience.

6.4. FINAL THOUGHTS:

In conclusion, the development of the Namal Blood Bank Management System has been a rewarding journey marked by collaborative effort, technical innovation, and valuable learning experiences.

Personally, this project has underscored the importance of perseverance and teamwork in overcoming challenges and achieving project goals. The system holds promise not only in streamlining blood bank operations but also in contributing to the broader healthcare ecosystem.

Acknowledgments are due to our project team, mentors, and supporters whose guidance and encouragement were instrumental throughout this endeavor.

As a team we look forward, the insights gained and the relationships forged during this project will undoubtedly shape our future endeavors in technology and healthcare.

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